

Beating a deployed rating system

Partial Pooling Against Per-Category Glicko-2 on Lichess

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Abstract

Lichess maintains a separate Glicko-2 rating for each time control, which is stratification: condition on the category, estimate within it, and accept the sparse-cell variance that follows. We compare it with a partially pooled alternative on 59,399 decisive games from one month of the site’s open database, in which a player’s ability is a level estimated from all of their games plus shrunk bullet and blitz offsets. The pooled rating is better by 0.0137 to 0.0182 nats of held-out log loss across three independent splits, every interval excluding zero, and roughly four fifths of that margin is the estimator, a batch fit against an online update, with one fifth the factor structure itself. Stratification helps Glicko-2 and hurts our scalar, because its benefit is capped by what pooling already captures while its cost is not. Opening family is unidentifiable because players choose openings, opponent-strength curvature is real but almost entirely shared, and modern Lichess no longer poses the question, with 0.3% of players mixing time controls against 52% in 2013.

1 Introduction

A rating is conventionally a number. Elo ([Elo, 1978](#)), Glicko ([Glickman, 1999](#)), TrueSkill ([Herbrich et al., 2007](#)) and their descendants all summarise a competitor by a scalar, optionally with an uncertainty attached. When a rating needs to reflect that ability varies by context, say a player strong at speed chess and ordinary at slow, the standard remedy is to stratify: keep one rating per category. Lichess does exactly this, maintaining independent Glicko-2 ratings for bullet, blitz and classical.

Stratification is a choice with a price. Each category’s rating is estimated only from that category’s games, so a player with many blitz games and few bullet games gets a well-estimated blitz rating and a noisy bullet one, and nothing the blitz games reveal about their general strength is allowed to inform the bullet number. The alternative is partial pooling: estimate the level from everything and let only the *deviations* be category-specific, shrunk toward zero. Ability becomes a vector $\mu_i(x) = d_i \cdot x$ with $x = [1, \mathbf{1}\{\text{bullet}\}, \mathbf{1}\{\text{blitz}\}]$, and a per-feature penalty leaves the level free while shrinking the offsets.

The question is which is better, measured against the deployed system rather than a textbook baseline, and what the difference is made of.

2 Data and protocol

One month of the Lichess open database (2013-01, CC0), reduced to headers. We keep decisive games between players with at least 100 games in the month, 59,399 games among 639 players;

draws are dropped. Each game is labelled with its time control.

Games are split 50/25/25 into fitting, validation and scored sets, and every arm sees the same games. Glicko-2’s τ is tuned on the validation split exactly as our ridge is; neither side is given a free parameter the other lacks. Held-out mean log loss is reported on the scored quarter only, and intervals come from a paired bootstrap over 20,000 resamples. Three independent splits are reported throughout, because a single split on this kind of data is not evidence.

Four arms are compared. *Glicko-2 pooled* ignores time control. *Glicko-2 per time control* is Lichess’s deployed architecture. *Our scalar* is a batch Thurstonian maximum-a-posteriori fit with one parameter per player, and exists to separate the estimator from the structure. *Our factor* adds the shrunk offsets.

3 Result

arm	seed 0	seed 11	seed 22
Glicko-2 pooled	0.6323	0.6304	0.6365
Glicko-2 per time control (<i>Lichess</i>)	0.6234	0.6241	0.6294
our scalar	0.6117	0.6131	0.6145
our factor	0.6073	0.6104	0.6112

The pooled factor rating beats the deployed architecture on every split. Decomposed:

comparison	seed 0	seed 11	seed 22
factor vs Lichess	−0.0161	−0.0137	−0.0182
estimator (scalar vs Glicko-2 pooled)	−0.0206	−0.0173	−0.0219
structure (factor vs our scalar)	−0.0044	−0.0027	−0.0033
stratification’s value to Glicko-2	+0.0089	+0.0063	+0.0071

Every interval excludes zero on every split, with $P(\text{better}) \geq 0.995$.

The headline should not be quoted without the decomposition. Roughly four fifths of the margin is the estimator, a batch fit that sees the training games at once against an online update that sees each game once, and one fifth is the factor structure. The structure’s own contribution, 0.003 to 0.004 with intervals excluding zero on all three splits, is the part that answers the question posed. The estimator column is a real advantage but a different claim, and it is partly a privilege: a production system must process games once, and ours need not.

4 Why stratification helps Glicko-2 and not the batch estimator

Two facts in the table look contradictory and are not. Stratifying Glicko-2 *gains* 0.0063–0.0089. Stratifying our scalar *loses* 0.0006. Partial pooling of the same information gains 0.0027–0.0044.

Stratification’s payoff runs inversely to the quality of the base estimator. Its benefit is the category information; its cost is sparse-cell variance. When the base estimator is weak, the benefit dominates. When the base estimator is strong, the pooled level already captures most of what the category would have told you, and only the variance cost remains. Partial pooling extracts the same signal without paying that cost, which is why it wins in both regimes and is the unconditional recommendation. A practitioner reporting that stratification helped them has reported something about their base estimator, not found a counterexample.

5 The covariates that did not work

The time-control result is not evidence that any covariate would do, and the failures are more informative than the success.

Opening family fails, and the mechanism generalises. A fair fit, with the offset ridge swept to the scalar limit, collapses exactly onto the scalar. The axis is nonetheless real: among 790 players with at least 20 games in each family, the style differential carries a true standard deviation of about 4.7 win-rate points, roughly four sigma above binomial noise. It is *unidentifiable, not absent*. Players choose their openings, so the within-player contrast that identification requires barely exists: the tactical share is bimodal with deciles 0.00/0.10/0.86/0.98/1.00 and a standard deviation of 0.415, against 0.058 under a no-choice null. A factor rating needs a covariate that varies within a player for reasons other than the player’s preference.

Opponent strength is real but shared. The signed opponent-strength gap passes the exogeneity requirement by construction, since pairing hands every player a spread of opponents. Fitted against a deliberately fair baseline, a scalar plus one global curvature coefficient rather than a bare scalar, the per-player offsets are identifiable, with a tuned ridge interior at 10.0 and surviving offset standard deviation 0.1316. But the gain is -0.00056 with interval $[-0.00117, +0.00005]$, spanning zero, and eight times smaller than time control. The global coefficient is -0.584 : one shared parameter absorbs nearly all of the opposition-strength signal, leaving only per-player deviations for the factor arm. Against a bare scalar this axis would have looked strong, and that comparison would have been a straw man.

So of three identifiable covariates, level, time control and opponent strength, the factor treatment pays on one.

6 Robustness

Colour is not the confound. Our arms model the white advantage and standard Glicko-2 does not, so the estimator column invites the suspicion that it is colour bookkeeping. It is not: colour is worth -0.0016 of the -0.0206 gap, and a colour-blind fit still beats colour-blind Glicko-2 by -0.0190 like for like.

The result is not an artifact of the dense subpopulation, but the mechanism’s own prediction failed. Rerunning at inclusion thresholds of 100/50/25/10 games moves observations per parameter from 30.9 to 15.3 and the margin stays between -0.016 and -0.019 throughout, every interval excluding zero. The robustness question is answered. The registered prediction, however, was that pooling should pay *most* where data is scarcest and the margin should widen. It does not: it is flat within noise and not monotone. The factor advantage is roughly constant in population density rather than concentrated among data-poor players, which leaves the pooling account unrefuted but unconfirmed by the one test that could have confirmed it.

Our arm is not winning by being overconfident. Glicko-2 propagates rating deviation into every prediction; our arm prices a point estimate with fixed noise, so only one side carries its own uncertainty. Giving ours a validation-tuned temperature returns $s^* = 1.1$ and improves held-out loss by 0.0001, with an interval spanning zero; the margin moves from -0.0161 to -0.0162 . The

arm was already calibrated and the asymmetry costs nothing. For completeness, pricing instead with a per-player predictive variance taken from the diagonal of the inverse penalised Hessian is significantly *worse*, by +0.0015.

7 Limitations

Batch against online. Our estimator revisits the training set; Glicko-2 processes each game once because a deployed system must. Part of the estimator column is that privilege, and the deployment-realistic comparison, online against online, is not run here.

Draws are dropped. They are 3.3% of games in this pool and far more at longer time controls and higher strength. A three-way outcome model is the right treatment.

One month of 2013. The obvious next step is a modern month, and it cannot currently be taken. In 2024-01 the time-control covariate is unidentifiable: only 0.3% of players have at least 20 games in two or more time controls, against 52% in 2013, at every inclusion threshold we tried. Contrast falls as activity rises, since the most active players are the most specialised, 93.1% bullet, so more data makes identification worse rather than better, and restricting to the players who do mix controls fails from the other side, leaving 904 players sharing 2,839 games at roughly one observation per parameter. This is not evidence against the result; no arm was fitted. It means the result holds on a population where players mixed time controls, and that population no longer exists at scale on this site.

8 Conclusion

A partially pooled vector rating beats a rating system in production, on its own data, by 0.0137 to 0.0182 nats. Most of that is the estimator and about a fifth is the pooling, and both parts are worth stating separately. The wider claim the data supports is narrower than “ratings should be vectors”: it is that *conditioning* is the expensive way to use category information and *pooling* is the cheap one, that the gap between them is largest for whoever has the weaker base estimator, and that the covariate must vary within a player for reasons the player does not choose. Opening choice fails that condition, opponent strength satisfies it while carrying almost no per-player signal, and time control satisfied it in 2013 and no longer does.

References

- A. E. Elo. *The Rating of Chessplayers, Past and Present*. Arco, New York, 1978.
- M. E. Glickman. Parameter estimation in large dynamic paired comparison experiments. *Journal of the Royal Statistical Society C* 48(3) (1999) 377–394.
- M. E. Glickman. Example of the Glicko-2 system. Boston University technical note, 2012.
- R. Herbrich, T. Minka, T. Graepel. TrueSkill(TM): a Bayesian skill rating system. *Advances in Neural Information Processing Systems* 19 (2007) 569–576.

- W. James, C. Stein. Estimation with quadratic loss. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability* 1 (1961) 361–379.
- B. Efron, C. Morris. Data analysis using Stein’s estimator and its generalizations. *Journal of the American Statistical Association* 70(350) (1975) 311–319.
- L. L. Thurstone. A law of comparative judgment. *Psychological Review* 34(4) (1927) 273–286.